

Отчёт по лабораторной работе 6

Адресация IPv4 и IPv6. Двойной стек

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Характеристика	Значение
Адрес сети	172.16.20.0/24
Префикс маски	/24
Маска	255.255.255.0
<u>Broadcast-адрес</u>	172.16.20.255/24
Адрес сети в двоичной форме	10101100.00010000.00010100.00000000
Маска в двоичной форме	11111111.11111111.11111111.00000000
Число возможных подсетей	$2^8 = 256$
Диапазон адресов узлов	172.16.20.1 - 172.16.20.254

Рис. 1: Топология лабораторного стенда в GNS3

Характеристика	Значение
Адрес сети	10.10.1.64/26
Префикс маски	/26
Маска	255.255.255.192
<u>Broadcast-адрес</u>	10.10.1.127/26
Адрес сети в двоичной форме	00001010.00001010.00000001.01000000
Маска в двоичной форме	11111111.11111111.11111111.11000000
Число возможных подсетей	$2^6 = 64$
Диапазон адресов узлов	10.10.1.65- 10.10.1.126

Рис. 2: Настройка IPv4-адресации на PC1-akabrikosov

Характеристика	Значение
Адрес сети	10.10.1.0/26
Префикс маски	/26
Маска	255.255.255.192
<u>Broadcast-адрес</u>	10.10.1.63/26
Адрес сети в двоичной форме	00001010.00001010.00000001.00000000
Маска в двоичной форме	11111111.11111111.11111111.11000000
Число возможных подсетей	$2^6 = 64$
Диапазон адресов узлов	10.10.1.1- 10.10.1.62

Рис. 3: Настройка IPv4-адресации на PC2-akabrikosov

Характеристика	Значение
Адрес сети	2a02:6b8::/64
Длина префикса	/64
Префикс	2a02:6b8:0:0
Маска	fff:fff:fff:fff:0000:0000:0000:0000
Диапазон адресов узлов	02a2:06b8:0000:0000:0000:0000:0000:0000- 02a2:06b8:0000:0000:fff:fff:fff:fff

Рис. 4: Настройка IPv4-адресации на Server-akabrikosov

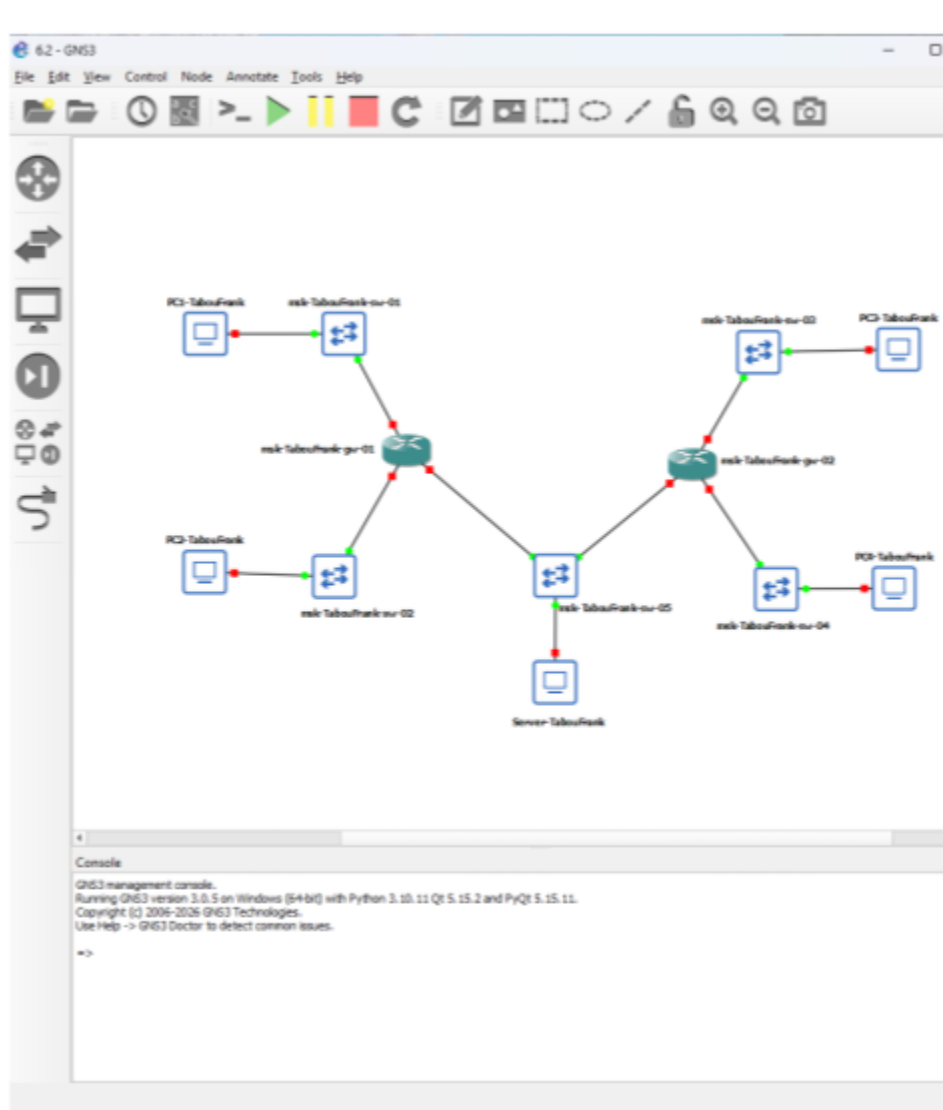


Рис. 5: Настройка интерфейсов маршрутизатора msk-akabrikosov-gw-01

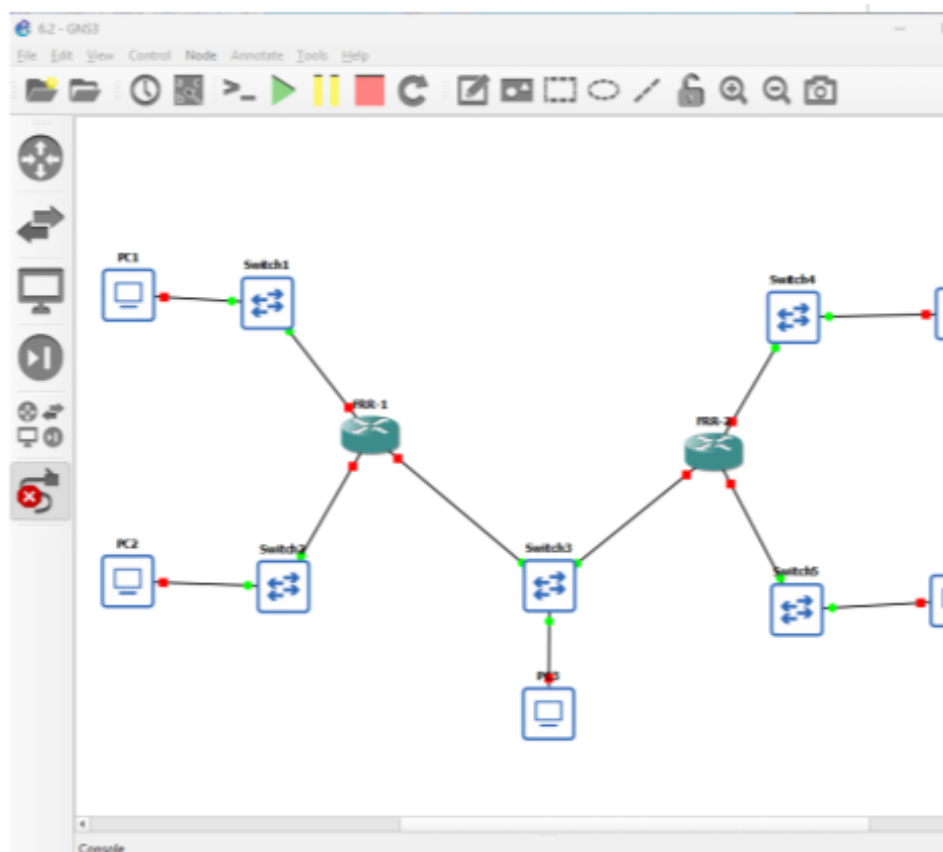


Рис. 6: Проверка состояния интерфейсов маршрутизатора

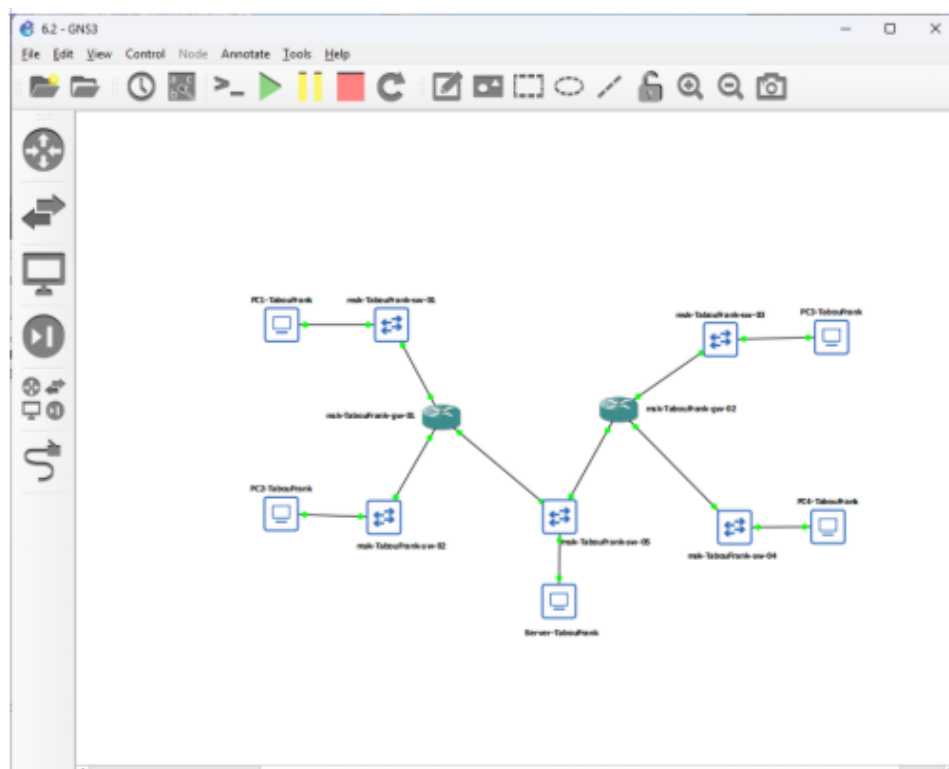


Рис. 7: Проверка связности с PC1-akabrikosov

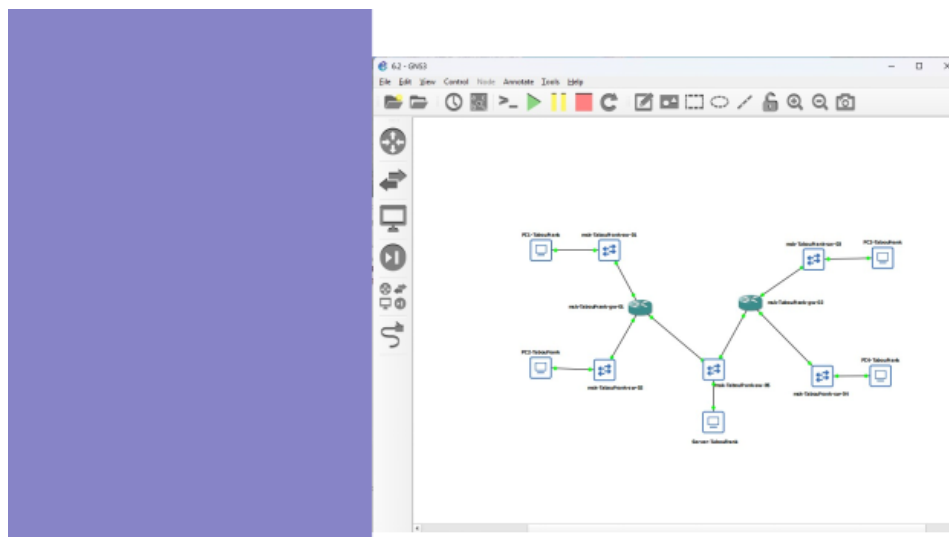


Рис. 8: Проверка связности с PC2-akabrikosov

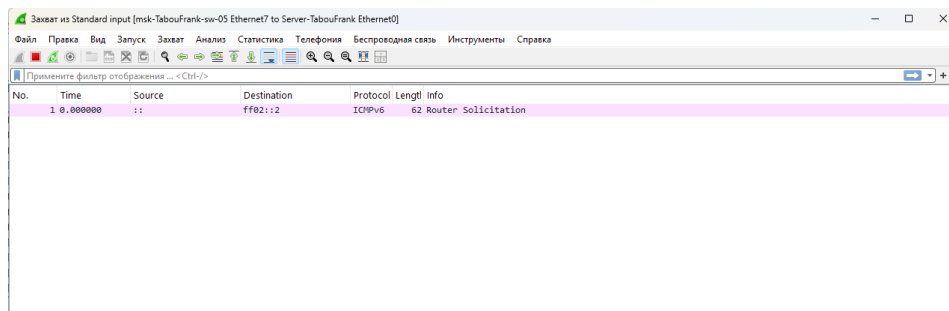


Рис. 9: Настройка IPv6-адресации на PC3-akabrikosov

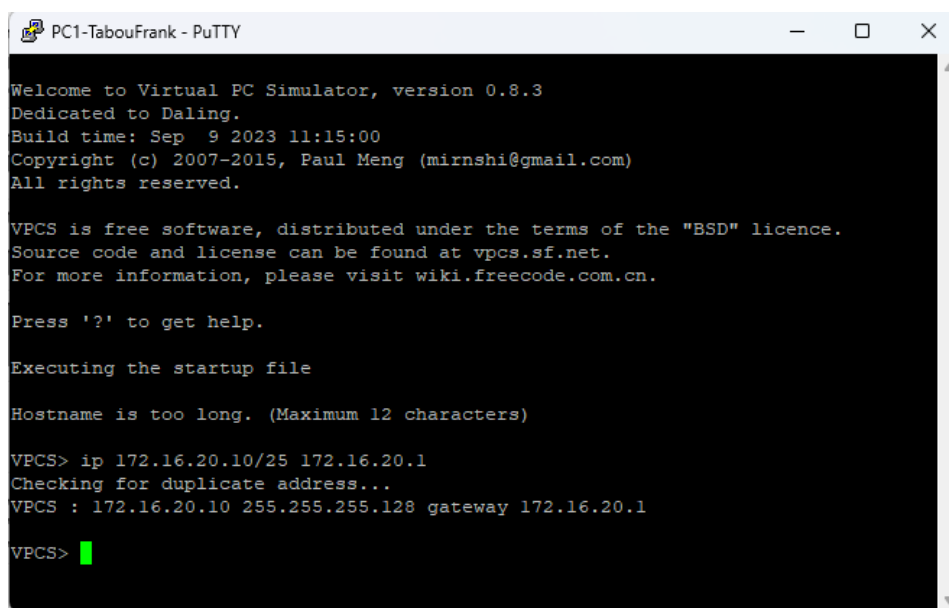


Рис. 10: Настройка IPv6-адресации на PC4-akabrikosov

```
PC1-TabouFrank - PuTTY

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 172.16.20.10/25 172.16.20.1
Checking for duplicate address...
VPCS : 172.16.20.10 255.255.255.128 gateway 172.16.20.1

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> sahow ip
Bad command: "sahow ip". Use ? for help.

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 172.16.20.10/25
GATEWAY    : 172.16.20.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT      : 10050
RHOST:PORT : 127.0.0.1:10051
MTU        : 1500

VPCS> show ipv6

NAME       : VPCS[1]
LINK-LOCAL SCOPE : fe80::250:79ff:fe66:6800/64
GLOBAL SCOPE    :
DNS             :
ROUTER LINK-LAYER :
MAC            : 00:50:79:66:68:00
LPORT          : 10050
RHOST:PORT     : 127.0.0.1:10051
MTU            : 1500

VPCS> 
```

Рис. 11: Проверка IPv4- и IPv6-адресации на Server-akabrikosov

```
PC2-TabouFrank - PuTTY
Welcome to Virtual PC Simulator, version 0.8.3
Dedicated to Daling.
Build time: Sep  9 2023 11:15:00
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 172.16.20.138/25 172.16.20.129
Checking for duplicate address...
VPCS : 172.16.20.138 255.255.255.128 gateway 172.16.20.129

VPCS> save
Saving startup configuration to startup.vpc
. done

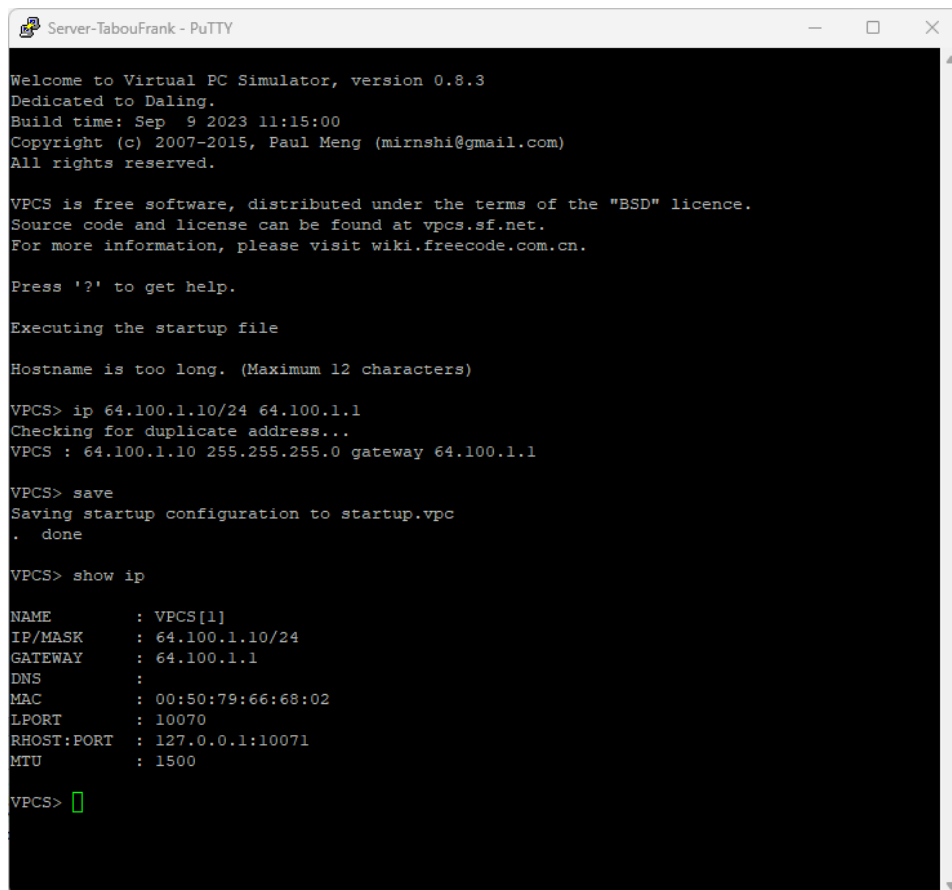
VPCS> shop ip
Bad command: "shop ip". Use ? for help.

VPCS> show ip

NAME      : VPCS[1]
IP/MASK    : 172.16.20.138/25
GATEWAY    : 172.16.20.129
DNS        :
MAC        : 00:50:79:66:68:01
LPORT     : 10052
RHOST:PORT : 127.0.0.1:10053
MTU        : 1500

VPCS> 
```

Рис. 12: Настройка IPv6-интерфейсов маршрутизатора VyOS



```
Server-TabouFrank - PuTTY

Welcome to Virtual PC Simulator, version 0.8.3
Dedicated to Daling.
Build time: Sep  9 2023 11:15:00
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 64.100.1.10/24 64.100.1.1
Checking for duplicate address...
VPCS : 64.100.1.10 255.255.255.0 gateway 64.100.1.1

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 64.100.1.10/24
GATEWAY    : 64.100.1.1
DNS        :
MAC        : 00:50:79:66:68:02
LPORT     : 10070
RHOST:PORT : 127.0.0.1:10071
MTU        : 1500

VPCS> 
```

Рис. 13: Проверка интерфейсов маршрутизатора VyOS

```
msk-TabouFrank-gw-01 - PuTTY
* Starting busybox syslog ... [ ok ]
* Seeding 256 bits and crediting
* Saving 256 bits of creditable seed for next boot
* Starting busybox acpid ... [ ok ]
* Starting busybox crond ... [ ok ]
Started watchfrr
* Starting sshd ... [ ok ]

Hello, this is FRRouting (version 8.2.2).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

frr# configure terminal
frr(config)# hostname msk-TabouFrank-gw-01
msk-TabouFrank-gw-01(config)# exit
msk-TabouFrank-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-TabouFrank-gw-01# configure terminal
msk-TabouFrank-gw-01(config)# interface eth0
msk-TabouFrank-gw-01(config-if)# ip address 172.16.20.1/25
% Unknown command: ip address 172.16.20.1/25
msk-TabouFrank-gw-01(config-if)# ip address 172.16.20.1/25
msk-TabouFrank-gw-01(config-if)# no shutdown
msk-TabouFrank-gw-01(config-if)# exit
msk-TabouFrank-gw-01(config)# interface eth1
msk-TabouFrank-gw-01(config-if)# ip address 172.16.20.129/25
msk-TabouFrank-gw-01(config-if)# no shutdown
msk-TabouFrank-gw-01(config-if)# exit
msk-TabouFrank-gw-01(config)# interface eth2
msk-TabouFrank-gw-01(config-if)# ip address 64.100.1.1/24
msk-TabouFrank-gw-01(config-if)# no shutdown
msk-TabouFrank-gw-01(config-if)# exit
msk-TabouFrank-gw-01(config)# exit
msk-TabouFrank-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-TabouFrank-gw-01# show interfaces
% Unknown command: show interfaces
msk-TabouFrank-gw-01# configure terminal
msk-TabouFrank-gw-01(config)# show interfaces
% Unknown command: show interfaces
msk-TabouFrank-gw-01(config)# exit
msk-TabouFrank-gw-01# show interfaces
% Unknown command: show interfaces
msk-TabouFrank-gw-01#
```

Рис. 14: Проверка IPv6-связности с РСЗ-akabrikosov

```

Commit failed
[edit]
vyos@vyos# delete interface ethernet eth0 hw-id
[edit]
vyos@vyos# commit

Can't configure both static IPv4 and DHCP address on the same interface

[[interfaces ethernet eth0]] failed
Commit failed
[edit]
vyos@vyos# compare
[edit interfaces ethernet eth0]
+address 172.16.20.1/25
-hw-id 0c:e5:a4:43:00:00
[edit]
vyos@vyos# set interfaces ethernet eth0 address 172.16.20.1/25

Configuration path: [interfaces ethernet eth0 address 172.16.20.1/25] already
exists
[edit]
vyos@vyos# set interfaces ethernet eth1 address 172.16.20.129/25

Configuration path: [interfaces ethernet eth1 address 172.16.20.129/25] alread
y exists
[edit]
vyos@vyos# set interfaces ethernet eth2 address 64.100.1.1/24

Configuration path: [interfaces ethernet eth2 address 64.100.1.1/24] already e
xists
[edit]
vyos@vyos# compare
[edit interfaces ethernet eth0]
+address 172.16.20.1/25
-hw-id 0c:e5:a4:43:00:00
[edit]
vyos@vyos# show interfaces
 ethernet eth0 {
     address dhcp
+   address 172.16.20.1/25
-   hw-id 0c:e5:a4:43:00:00
 }
 ethernet eth1 {
     address 172.16.20.129/25
     hw-id 0c:e5:a4:43:00:01
 }
 ethernet eth2 {
     address 64.100.1.1/24
     hw-id 0c:e5:a4:43:00:02
 }
 loopback lo {
 }
[edit]
vyos@vyos# creat interface ethernet eth0 hw-id

Invalid command: [creat]
[edit]
vyos@vyos# show interfaces

```

Рис. 15: Проверка IPv6-связности с PC4-akabrikosov

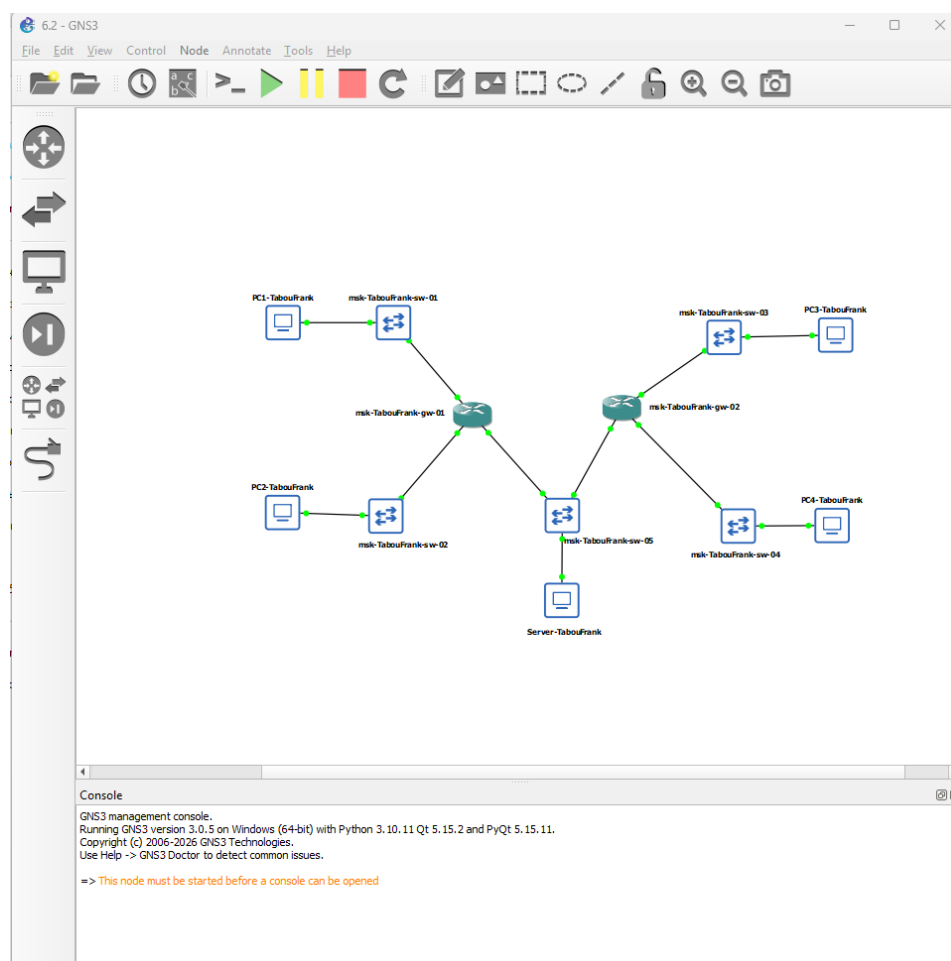


Рис. 16: Проверка изоляции IPv4 и IPv6 подсетей

```
msk-TabouFrank-gw-01 - PuTTY
* Starting busybox syslog ... [ ok ]
* Seeding 256 bits and crediting
* Saving 256 bits of creditable seed for next boot
* Starting busybox acpid ... [ ok ]
* Starting busybox crond ... [ ok ]
Started watchfrr
* Starting sshd ... [ ok ]

Hello, this is FRRouting (version 8.2.2).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

frr# configure terminal
frr(config)# hostname msk-TabouFrank-gw-01
msk-TabouFrank-gw-01(config)# exit
msk-TabouFrank-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-TabouFrank-gw-01# configure terminal
msk-TabouFrank-gw-01(config)# interface eth0
msk-TabouFrank-gw-01(config-if)# ip address 172.16.20.1/25
% Unknown command: ip adress 172.16.20.1/25
msk-TabouFrank-gw-01(config-if)# ip address 172.16.20.1/25
msk-TabouFrank-gw-01(config-if)# no shutdown
msk-TabouFrank-gw-01(config-if)# exit
msk-TabouFrank-gw-01(config)# interface eth1
msk-TabouFrank-gw-01(config-if)# ip address 172.16.20.129/25
msk-TabouFrank-gw-01(config-if)# no shutdown
msk-TabouFrank-gw-01(config-if)# exit
msk-TabouFrank-gw-01(config)# interface eth2
msk-TabouFrank-gw-01(config-if)# ip address 64.100.1.1/24
msk-TabouFrank-gw-01(config-if)# no shutdown
msk-TabouFrank-gw-01(config-if)# exit
msk-TabouFrank-gw-01(config)# exit
msk-TabouFrank-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-TabouFrank-gw-01# show interfaces
% Unknown command: show interfaces
msk-TabouFrank-gw-01# configure terminal
msk-TabouFrank-gw-01(config)# show interfaces
% Unknown command: show interfaces
msk-TabouFrank-gw-01(config)# exit
msk-TabouFrank-gw-01# show interfaces
% Unknown command: show interfaces
msk-TabouFrank-gw-01#
```

Рис. 17: Захват ARP-трафика на интерфейсе сервера


```

Commit failed
[edit]
vyos@vyos# delete interface ethernet eth0 hw-id
[edit]
vyos@vyos# commit

Can't configure both static IPv4 and DHCP address on the same interface

[[interfaces ethernet eth0]] failed
Commit failed
[edit]
vyos@vyos# compare
[edit interfaces ethernet eth0]
+address 172.16.20.1/25
-hw-id 0c:e5:a4:43:00:00
[edit]
vyos@vyos# set interfaces ethernet eth0 address 172.16.20.1/25

Configuration path: [interfaces ethernet eth0 address 172.16.20.1/25] already exists
[edit]
vyos@vyos# set interfaces ethernet eth1 address 172.16.20.129/25

Configuration path: [interfaces ethernet eth1 address 172.16.20.129/25] already exists
[edit]
vyos@vyos# set interfaces ethernet eth2 address 64.100.1.1/24

Configuration path: [interfaces ethernet eth2 address 64.100.1.1/24] already exists
[edit]
vyos@vyos# compare
[edit interfaces ethernet eth0]
+address 172.16.20.1/25
-hw-id 0c:e5:a4:43:00:00
[edit]
vyos@vyos# show interfaces
ethernet eth0 {
    address dhcp
+   address 172.16.20.1/25
-   hw-id 0c:e5:a4:43:00:00
}
ethernet eth1 {
    address 172.16.20.129/25
    hw-id 0c:e5:a4:43:00:01
}
ethernet eth2 {
    address 64.100.1.1/24
    hw-id 0c:e5:a4:43:00:02
}
loopback lo {
}
[edit]
vyos@vyos# creat interface ethernet eth0 hw-id

Invalid command: [creat]
[edit]
vyos@vyos# show interfaces

```

Рис. 18: Захват ICMP-трафика IPv4

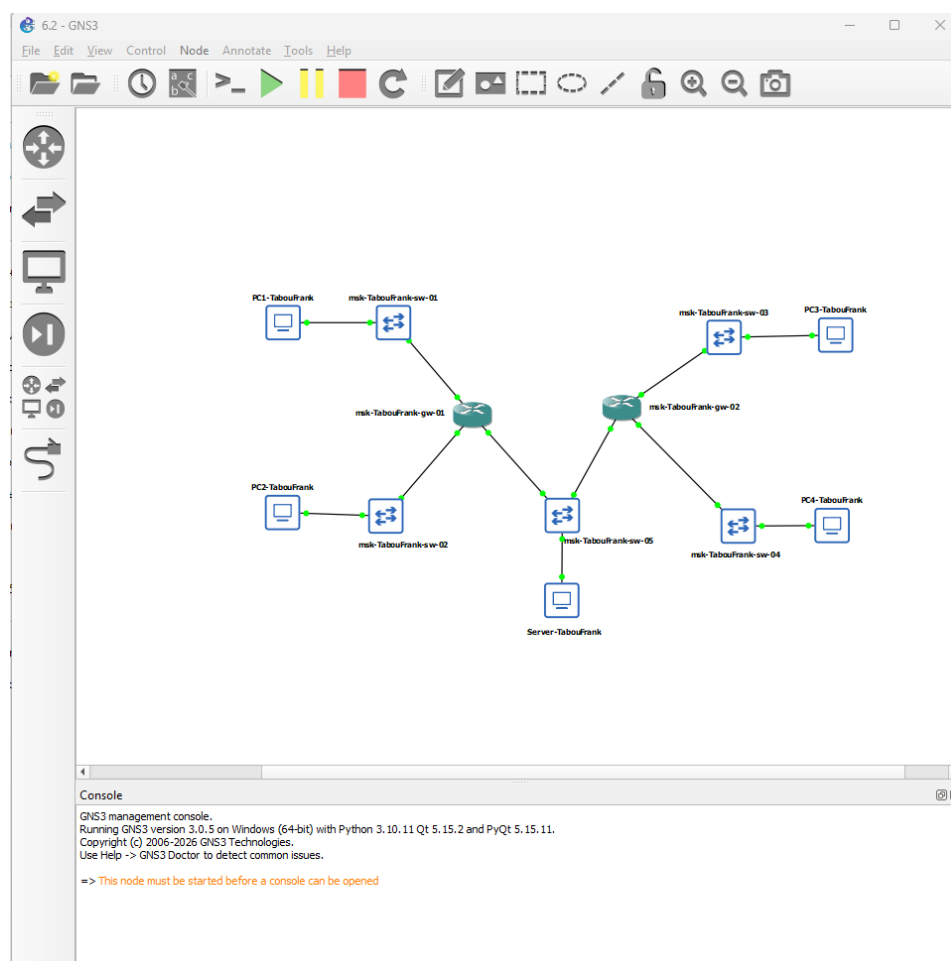


Рис. 19: Захват ICMPv6-трафика